# SEONGGYU PARK

Dr. Seonggyu Park's research focuses on the development of hydrologic models for identifying challenges in the present and future water resources management.

Recent projects include:

- Assessment of spatio-temporal patterns of surface and subsurface water resources, interactions, and water quality
- Assessment of water, salt, fire, and land management
- Investigating impacts of groundwater abstraction, climate changes, and land management on water recourses


## CONTACT

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View this short CV online with link at spark-brc.github.io/short_cv
*Last updated on December 8th, 2020*

## PROJECTS

Current | Apr. 2020

### Assessment of Water, Salt, Fire and Land Management in the Colorado River Basin (CRB) using APEX

Texas A&M AgriLife Research, Bureau of Land Management

- Assess Spatio-Temporal Patterns of Groundwater - Surface Water Interactions and Solute Transports in the Colorado River Basin (CRB) Using a Coupled APEX-MODLOW Model: Animas, White, Price, Green, Doleres River Watersheds APEX-MODFLOW
- Assess Water, Salt, Fire and Land Mangement in the Colorado River Basin using APEX APEX
- Tool Development: APEXMOD https://github.com/spark-brc/APEXMOD Python Qt

Current | Apr. 2020

### SWAT-MODFLOW modeling to support resilient development and return on investment analyses in the Cubango-Okavango River Basin (CORB)

Texas A&M AgriLife Research, The Nature Conservancy, National Geographic

- Couple the existing SWAT (Soil and Water Assessment Tool) model with USGS MODFLOW to more accurately model sub-surface hydrology SWAT-MODFLOW
- Optimize SWAT-MODFLOW using Parameter ESTimation Tool PEST Python
- Info: Okvavango River Basin

Mar. 2020 | Nov. 2018

### Conservation Effects and Assessment Project (CEAP) - National SWAT+ model assessments

Texas A&M AgriLife Research, U.S. Department of Agriculture Agricultural Research Service

- Provided technical supports by developing database, framework, and analysis tools for the Real-Time Conservation Effects and Assessment Project (CEAP)
- Interactive web-based dashboard: https://github.com/spark-brc/NM-viz Python Jupyter Notebook
  Tutorial Video: https://www.youtube.com/watch?v=9OLHA5hhEek

## PUBLICATIONS

In preparation

**APEX-CUTE: A Graphical User Interface for Senstivity Analysis and Parameter Estimation of Agricultural Policy and Environmental Extender (APEX) model**

*PLOS*

• **Park, S.**, Jeong, J., Abitew, T.

In preparation

**Parameter estimation and undertainty analysis framework for integrated surface-groundwater modeling**

*Hydrological Processes*

• **Park, S.**, Jeong, J., Abitew, T.

In preparation

**SWAT-MODFLOW modeling to support resilient development and return on investment analyses in the Cubango-Okavango River Basin (CORB)**

*Science of Total Environment*

• **Park, S.**, Jeong, J., Abitew, T., Baker, T., …

In preparation

**Parameter Estimation Approaches for Coupled Surface / Subsurface Models**

*Geoscientific Model Development*

• **Park, S.**, Bailey, R, James, S., Schürz, C., Arnold, J.

In preparation

**APEX-MODFLOW: A New Integrated Watershed Model to Simulate Water Movement at the Watershed Scale**

*Environmental Modelling & Software*

• Bailey, R.T., Tavakoli, S., Tasdighi, A., Jeong, J., **Park, S.**

2020

**Quantifying the effects of climate change on hydrological regime and stream biota in a groundwater-dominated catchment: A modelling approach combining SWAT-MODFLOW with flow-biota empirical models**

*Science of The Total Environment*, 2020, 745, p.140933. DOI 10.1016/j.scitotenv.2020.140933

• Liu, W., Bailey, R.T., Andersen, H.E., Jeppesen, E., Nielsen, A., Peng, K., Molina-Navarro, E., **Park, S.**, Thodsen, H. and Trolle, D.

2020

**Enhancing SWAT+ simulation of groundwater flow and groundwater-surface water interactions using MODFLOW routines**

*Environmental Modelling & Software*, 2020, 126, p.104660. DOI 10.1016/j.envsoft.2020.104660

• Bailey, R.T., **Park, S.**, Bieger, K., Arnold, J.G. and Allen, P.M.

2020

**Assessing the impacts of groundwater abstractions on flow regime and stream biota: Combining SWAT-MODFLOW with flow-biota empirical models**

*Science of the Total Environment*, 2020, 706, p.135702. DOI 10.1016/j.scitotenv.2019.135702

• Liu, W., Bailey, R.T., Andersen, H.E., Jeppesen, E., **Park, S.**, Thodsen, H., Nielsen, A., Molina-Navarro, E. and Trolle, D

2019

**Modelling potential groundwater recharge in the Limpopo River Basin with SWAT-MODFLOW**

*Groundwater for Sustainable Development*, 2019, 9, p.100260. DOI 10.1016/j.gsd.2019.100260

• Mosase, E., Ahiablame, L., **Park, S.** and Bailey, R.

2019

**IPEAT+: a Built-in optimization and automatic calibration tool of SWAT+**

*Water*, 2019, 11(8), p.1681.

DOI 10.3390/w11081681

• Yen, H., **Park, S.**, Arnold, J.G., Srinivasan, R., Chawanda, C.J., Wang, R., Feng, Q., Wu, J., Miao, C., Bieger, K. and Daggupati, P.

2019

**Comparing SWAT with SWAT-MODFLOW hydrological simulations when assessing the impacts of groundwater abstractions for irrigation and drinking water**

*Hydrol. Earth Syst. Sci. Discuss*, 2019, pp.1-51.

DOI 10.5194/hess-2019-232

• Liu, W., **Park, S.**, Bailey, R.T., Molina-Navarro, E., Andersen, H.E., Thodsen, H., Nielsen, A., Jeppesen, E., Jensen, J.S., Jensen, J.B. and Trolle, D.

2019

**Comparison of abstraction scenarios simulated by SWAT and SWAT-MODFLOW**

*Hydrological Sciences Journal*, 2019, 64(4), pp.434-454.

DOI 10.1080/02626667.2019.1590583

• Molina-Navarro, E., Bailey, R.T., Andersen, H.E., Thodsen, H., Nielsen, A., **Park, S.**, Jensen, J.S., Jensen, J.B. and Trolle, D.

2019

**A QGIS-based graphical user interface for application and evaluation of SWAT-MODFLOW models**

*Environmental Modelling & Software*, 2019, 111, pp.493-497.

DOI 10.1016/j.envsoft.2018.10.017

• **Park, S.**, Nielsen, A., Bailey, R.T., Trolle, D. and Bieger, K.